AIP Journal of Applied Physics

Plasmonic formation mechanism of periodic 100-nm-structures upon femtosecond laser irradiation of silicon in water

T. J.-Y. Derrien, R. Koter, J. Krüger, S. Höhm, A. Rosenfeld, and J. Bonse

Citation: Journal of Applied Physics **116**, 074902 (2014); doi: 10.1063/1.4887808
View online: http://dx.doi.org/10.1063/1.4887808
View Table of Contents: http://scitation.aip.org/content/aip/journal/jap/116/7?ver=pdfcov
Published by the AIP Publishing

JOURNAL OF APPLIED PHYSICS **116**, 074902 (2014)

Plasmonic formation mechanism of periodic 100-nm-structures upon femtosecond laser irradiation of silicon in water

T. J.-Y. Derrien,[1,a)] R. Koter,[1] J. Krüger,[1] S. Höhm,[2] A. Rosenfeld,[2] and J. Bonse[1,b)]
[1]*BAM Bundesanstalt für Materialforschung und –prüfung, Unter den Eichen 87, D-12205 Berlin, Germany*
[2]*Max-Born-Institut für Nichtlineare Optik und Kurzzeitspektroskopie (MBI), Max-Born-Staße 2A, D-12489 Berlin, Germany*

(Received 16 April 2014; accepted 26 June 2014; published online 15 August 2014)

The formation of laser-induced periodic surface structures (LIPSS) upon irradiation of silicon by multiple ($N = 100$) linearly polarized Ti:sapphire femtosecond laser pulses (duration $\tau = 30$ fs, center wavelength $\lambda_0 \sim 790$ nm) is studied experimentally in air and water environment. The LIPSS surface morphologies are characterized by scanning electron microscopy and their spatial periods are quantified by two-dimensional Fourier analyses. It is demonstrated that the irradiation environment significantly influences the periodicity of the LIPSS. In air, so-called low-spatial frequency LIPSS (LSFL) were found with periods somewhat smaller than the laser wavelength ($\Lambda_{LSFL} \sim 0.7 \times \lambda_0$) and an orientation perpendicular to the laser polarization. In contrast, for laser processing in water a reduced ablation threshold and LIPSS with approximately five times smaller periods $\Lambda_{LIPSS} \sim 0.15 \times \lambda_0$ were observed in the same direction as in air. The results are discussed within the frame of recent LIPSS theories and complemented by a thin film based surface plasmon polariton model, which successfully describes the tremendously reduced LIPSS periods in water. [http://dx.doi.org/10.1063/1.4887808]

I. INTRODUCTION

Modern technology and industrial applications generate an increasing demand of surface nanostructuring and functionalization with simple processing strategies and for reduced structural sizes. One promising way is based on laser-induced periodic surface structures (LIPSS), which are formed in a single-step process upon irradiation of solids by linearly polarized laser pulses. LIPSS have been observed first by Birnbaum in 1965 after irradiation of Germanium surfaces by pulsed ruby laser radiation.[1] For strong absorbing materials such as metals or semiconductors, the surface structures usually have spatial periods close to the laser wavelength. Typically, they are referred to as *low-spatial frequency LIPSS* (LSFL).

Using ultrashort laser pulses in the femtosecond (fs) domain, LIPSS with periods deviating significantly from the laser wavelength were observed. A period of about 500 nm was observed for 248-nm 500-fs laser pulse treatment of fused silica,[2] while 200 nm periodicity was noticed for 620-nm 300-fs irradiation of the same material.[3] LIPSS with periods smaller than half of the laser wavelength are called *high-spatial frequency LIPSS* (HSFL). They are observed predominantly after the irradiation of dielectrics or semiconductors with ultrashort laser pulses of fs- to ps-duration.[4,5]

One reason for that lies in the minimized *heat-affected zone* (HAZ) manifested in fs-laser processing when compared to longer pulse durations. For ns-laser pulses already during the pulse the absorbed optical energy spreads out from the excited volume via thermal diffusion, resulting in a heat-affected zone typically in the micrometer range.[6] For fs-laser pulse irradiation, the HAZ is much smaller and can account for a few hundred of nanometers only.[7]

In other words, a prerequisite for the observation of HSFL is the lateral confinement of the optical energy deposited into the solids (HAZ $\leq \Lambda_{HSFL}$). This requirement is typically facilitated by materials having a low carrier mobility, such as dielectrics. This idea of energy confinement further supports HSFL formation models that are based on ultrafast excitation mechanisms, such as radiation remnants,[9] second harmonic generation,[10] or plasmonic excitations.[11]

Another way to significantly reduce the LIPSS periods is a change to a liquid processing environment.[12–14] In this work, we investigate the environmental influence (air vs. water) on LIPSS formed on the technologically relevant material silicon. An extended plasmonic thin-film model is presented which includes the sample environment, seamless adopts the LSFL theory for irradiation of silicon in air and quantitatively explains the strongly reduced LIPSS periods observed for processing in water.

II. EXPERIMENTAL

Standard single-crystalline silicon wafers were selected as semiconducting material [n-doped, (111)-oriented, single-side polished, 400 μm thick, Werk für Fernsehelektronik, Berlin, Germany].

The samples were irradiated by multiple (N) femtosecond laser pulses either in air or in distilled water environment. Figure 1 presents the experimental setup used for fs-laser irradiation in a water-filled cell with an optical entrance window (fused silica, a-SiO_2). In that cell, the beam propagation distance between the window and the front surface of the sample is ~5 mm. For irradiation in air, the liquid

[a)]Electronic mail: thibault.derrien@gmail.com
[b)]Electronic mail: joern.bonse@bam.de

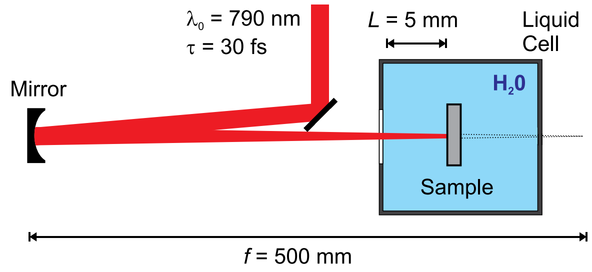

FIG. 1. Experimental setup for fs-laser processing of strong absorbing materials in water environment. f: focal length. L: distance between cell window and front surface of the sample.

containing cell was removed and the sample was exposed to the beam directly.

A chirped pulse Ti:sapphire laser amplifier system (Femtolasers, Compact Pro) was used to generate linearly polarized laser pulses of $\tau = 30$ fs duration at $\lambda_0 = 790$ nm center wavelength (1.6 eV single photon energy) at pulse repetition frequencies between $\nu = 1$ Hz (processing in water) and 1 kHz (processing in air). The laser pulses were focused by a spherical dielectric mirror ($f = 500$ mm focal length) and the samples were placed in front of the geometrical focus position in order to reduce nonlinear optical effects in air or water environment. At the sample surface, Gaussian beam radii $w_0(1/e^2)$ of $\sim$140 μm were obtained here.

The laser pulse energies were measured with a pyroelectric detector (Coherent 3Σ with J-25LP-3A-2K head) in front of the sample (air) or in front of the liquid cell (water). The peak fluences of the Gaussian-like beam profile were determined according to a method proposed by Liu.[15] Consequently, in the following, ϕ_0 represents either the peak fluence in front of the sample (air) or in front of the cell (water). All irradiations using $N = 100$ pulses per spot were performed under normal incidence to the sample.

The laser pulse irradiated surface regions were characterized by scanning electron microscopy (SEM, Carl Zeiss Gemini Supra 40). At selected spots, energy dispersive X-ray analyses (EDX) were performed in an elemental mapping mode using a Bruker Quantax 400 System. The mappings were performed at 3 kV electron acceleration voltage of the primary electrons in order to ensure the necessary surface sensitivity.

III. RESULTS AND DISCUSSION

For silicon, the formation of fs-LIPSS has been studied already in air[16–18] and water[12,17,18] environment. However, the excitation conditions do widely vary among those studies. Hence, we have performed a comparative study for similar excitation conditions (τ, λ_0, N).

A. Ablation thresholds for laser processing in air and water environment: The role of refractive index matching

In a first set of experiments, the damage characteristic of the materials was tested in both environments by variation of the fs-laser pulse energy and subsequent sample inspection. It was observed that generally, the damage threshold in water is lower by some tens of percent compared to air environment—in good agreement with the work of Wang *et al.*[17] Moreover, in water, the damage starts at specific damage locations and not necessarily in the center of the irradiated spot, where the highest local fluence should be expected. One possible origin is a transient (intrapulse) defocusing and beam filamentation due to the Kerr-effect in water as its nonlinear refractive index n_2 is approximately three orders of magnitude larger than that of air [n_2(water) $\sim 2.8 \times 10^{-16}$ cm^2/W, n_2(air) $\sim 2.1 \times 10^{-19}$ cm^2/W].[19] Additionally, the heterogeneous damage behaviour in water is affected by the generation of laser-induced gas bubbles at the sample/liquid interface. Consequently, the experiments in water were performed at 1 Hz pulse repetition frequency only in order to allow microscopic bubbles formed in the water to escape from the irradiated region.

For a more detailed analysis of the damage threshold fluences, the (Fresnel-) reflection losses at all interfaces of the liquid cell have to be taken into consideration. For normal incident radiation, the optical transmission through the interface between medium p and medium q (both assumed to be non-absorbing) can be calculated according to

$$T_{pq} = 1 - R_{pq} \tag{1}$$

from the reflectivity R_{pq} of the interface

$$R_{pq} = \left| \frac{\tilde{n}_p - \tilde{n}_q}{\tilde{n}_p + \tilde{n}_q} \right|^2. \tag{2}$$

Here, $\tilde{n}_p$ and $\tilde{n}_q$ represent the complex refractive index of the medium p or q, respectively. Hence, the laser pulse energy incident to our liquid cell has to be multiplied by a factor which accounts for reflection losses at all interfaces along the optical path of the laser beam, i.e., air/silica, silica/water, and water/sample.

For processing in water, the fluence ϕ_{abs} absorbed in the (opaque) sample material can then be calculated to be the incident fluence ϕ_0 multiplied with the transmission factors of all interfaces $T_{tot} := \prod_{p=0}^{p=2} T_{p,q=p+1}$ via

$$\phi_{abs} = \phi_0 \times T_{tot} = \phi_0 \times (1 - R_{air/silica}) \times (1 - R_{silica/water}) \times (1 - R_{water/sample}). \tag{3}$$

For processing in air (without using the liquid cell), the absorbed fluence simply accounts to

$$\phi_{abs} = \phi_0 \times T_{tot} = \phi_0 \times (1 - R_{air/sample}). \tag{4}$$

For evaluation of Eqs. (3) and (4), Table I compiles the (complex) refractive indices $\tilde{n} = n + ik$ of all materials relevant in the present study (n: refractive index, k: extinction coefficient).

Table II summarizes ablation threshold fluences ϕ_{th} of silicon experimentally determined for 100 pulses in air and water environment. Additionally, the overall transmission $T_{tot} = \phi_{abs}/\phi_0$ calculated from Eq. (3) or (4) is given for both cases. Multiplying these two values, the fluence absorbed at the ablation threshold within the sample material $\phi_{abs,th}$ can be estimated and is listed in the right column of the table.

TABLE I. Optical constants (complex refractive index, $n+ik$) of materials relevant in this study at the laser irradiation wavelength of $\lambda_0 \sim 790$ nm.

Material	n	k	Reference
Air	1.000	0	19
Water	1.326	0	20
Fused silica	1.453	0	20
Silicon	3.693	0.006	20

Generally, the experimental ablation threshold ϕ_{th} in water is lowered by a few tens of percent when compared to air environment.[13,17] For a fixed pulse duration of 100 fs, Miyaji *et al.* reported a ratio $\phi_{th}(\text{water})/\phi_{th}(\text{air})$ of 70% for silicon.[13] In our case here (30 fs), the reduction accounts to ~75% which can be mainly attributed to beam breakup and filamentation due to Kerr effect, generating hot spots in the spatial beam profile upon propagation in the water.

One reason for the threshold reduction lies in the refractive index matching of the sample surrounding medium (environment) and the liquid containing cell. This can be seen from the overall transmission of the fs-laser radiation into the sample material. For silicon, the calculated ratio $T_{tot}(\text{air})/T_{tot}(\text{water})$ is $0.67/0.75 \sim 90\%$ (Table II). These values are in close agreement with the ones reported by Wang *et al.* for fs-laser irradiation of silicon (82%) in air and water.[17]

B. Surface morphology (SEM)

Figure 2 compares the fs-laser processing of silicon surfaces in air [(a) + (c)] and in water environment [(b) + (d)] for a fixed number of $N = 100$ laser pulses per spot [air: $\phi_0 = 0.17\,\text{J/cm}^2$, water: $\phi_0 = 0.09\,\text{J/cm}^2$]. The fluence was individually optimized for each environment, resulting in the most pronounced LIPSS. Note that the most regular LIPSS are formed in air at a lower fluence ratio $\phi_0/\phi_{th} \sim 1.3$ than in water, where it is ~2.9. The upper row of Fig. 2 shows SEM images of surface regions near the center of the irradiation spot, while the lower row provides the corresponding two-dimensional Fourier transform (2D-FT) images.

For laser processing of silicon in air, LSFL with periods between 520 nm and 620 nm are directed perpendicular to the polarization [Figs. 2(a) and 2(c)]. These structures are caused by the excitation of surface plasmon polaritons (SPP) at the rough air/sample surface when the semiconducting silicon surface turns into a metallic state due to the generation of electrons in the conduction band of the solid.[8,21,22] The interference of the SPP electromagnetic field with the fs-laser pulse spatially modulates the optical energy deposited to the material and leads, after melting and ablation, to a corrugated surface relief. In contrast, for laser processing in water, significantly smaller LIPSS with periods between 80 and 130 nm are visible in Figs. 2(b) and 2(d) always perpendicular to the polarization direction.

TABLE II. Values of the experimental ablation threshold fluence ϕ_{th} ($N = 100$), the calculated total transmission $T_{tot} = \phi_{abs}/\phi_0$, and the estimated absorbed fluence threshold $\phi_{abs,th}$ for silicon irradiated in air and in water environment.

Material	Environment	ϕ_{th} (J/cm^2)	T_{tot}	$\phi_{abs,th}$ (J/cm^2)
Silicon	Air	0.13	0.67	0.087
	Water	0.03	0.75	0.022

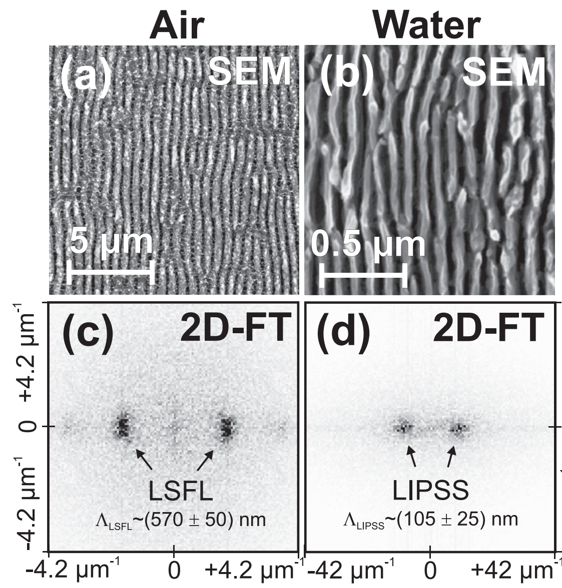

FIG. 2. SEM micrographs [(a) + (b)] and corresponding 2D-FT [(c) + (d)] of a silicon surface after irradiation with $N = 100$ fs-laser pulses in air [(a) + (c): $\phi_0/\phi_{th} \sim 1.3$] and in water environment [(b) + (d): $\phi_0/\phi_{th} \sim 2.9$]. Note the different scale bars in (a) and (b). The laser beam polarization is horizontal.

C. Chemical analysis (EDX)

In order to characterize chemical alterations induced at the rippled surface area upon fs-laser irradiation in air and water, a mapping of the spatial distributions of the elements silicon (Si), oxygen (O), and carbon (C) was performed in EDX analyses. The upper part of Fig. 3 shows SEM

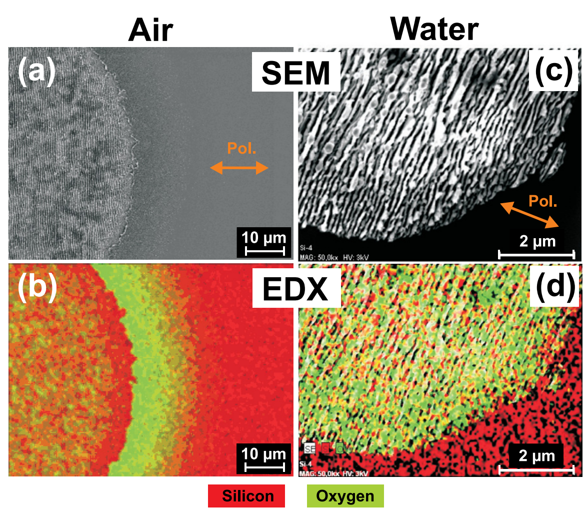

FIG. 3. SEM images of a LIPSS-covered area upon fs-laser processing of silicon in air [(a): $\phi_0/\phi_{th} \sim 1.3$, $N = 100$] and in water [(b): $\phi_0/\phi_{th} \sim 2.9$, $N = 100$]. In (b) and (d), the EDX elemental distributions of silicon (red) and oxygen (green) are superimposed to the corresponding SEM images. The laser beam polarization is indicated by arrows in (a) and (c). Note the different magnifications for processing in air and water.

micrographs of the border of a LIPSS covered area processed in air (a) and in water (c). In the lower part of Fig. 3, these images have been superimposed with the accumulated EDX signals of silicon (X-ray intensity at 1.74 keV encoded in red) and of oxygen (X-ray intensity at 0.53 keV encoded in green) for the processing in air (b) and water (d). As the carbon EDX signal did not show any significant correlation with the laser-generated surface morphology in both cases, its visualization is suppressed here for simplicity.

At the given conditions, in both processing environments, the elemental distributions of Si and O did not show a direct correlation with the LIPSS, e.g., individual ripple lines of the surface morphology. However, in both LIPSS-covered (ablated) areas (air and water), the O signal is increased compared to the non-irradiated regions, while the Si signal is decreased, indicating a superficial oxidation upon fs-laser processing in the oxygen containing environments. Such a laser-induced oxidation was reported already for the irradiation of silicon upon ns-laser irradiation in air.[23] The most striking difference between the two environmental cases (air and water) here is the observation of a $\sim$10 μm wide annular region around the ablation crater with a strongly increased amount of oxygen after the fs-laser irradiation in air [Fig. 3(b)]. It is associated with a thin layer of redeposited material (debris) which oxidized during the ablation in air. Since the transport of ablation products is strongly suppressed in the confining water environment, this annular feature is absent then [Fig. 3(d)]. Influences of differences in the pulse repetition frequency (air: 1 kHz/water: 1 Hz) are negligible here as in both cases the surface cools down to room temperature between successive pulses and heat accumulation effects can be excluded. Similarly, the peak laser fluence is not the decisive experimental parameter here as the oxidation effect in air is stronger than in water although the peak fluence ratio is smaller [air: $\phi_0/\phi_{th} \sim 1.3$; water: $\phi_0/\phi_{th} \sim 2.9$].

D. Comparison to literature data and existing theories

Table III compares the experimental results of the previous section to the relevant available literature. For each environmental condition, the data are ordered in a list of increasing periods Λ_{LIPSS}. The literature survey is limited to studies that reported LIPSS at low pulse repetition frequencies ($\nu \leq 1$ kHz) and without any additional surface post-treatment such as chemical etching.

TABLE III. Literature survey of LIPSS characteristics reported on single-crystalline silicon (c-Si) surfaces ($\lambda = 740$ to 800 nm, $\tau = 30$ to 160 fs, $\nu \leq 1$ kHz) under nearly normal incidence in air/vacuum or water environment. LIPSS orientation: $\perp$—ripples aligned perpendicular to polarization; $\|$—ripples aligned parallel to polarization. n.s.: not specified.

Material	Environment	Λ_{LIPSS} (nm)	Orientation/Type	Ref.
Silicon	Air/vac.	200	$\|$/HSFL	24
		520–620	$\perp$/LSFL	This work
		560–770	$\perp$/LSFL	8,12, and 21
		600	$\perp$/LSFL	13
		600–700	$\perp$/LSFL	24
		670	$\perp$/LSFL	17
	Water	80–130	$\perp$	This work
		100	n.s.	12
		120	$\perp$	17
		130–500	$\perp$	13

In air and under similar excitation conditions, LSFL were found with periods somewhat smaller than the laser wavelength [$\Lambda_{LSFL} \sim 0.7 \times \lambda_0$] and with an orientation perpendicular to the laser polarization. In contrast, for irradiation in water, reduced damage thresholds and LIPSS with approximately five times smaller periods $\Lambda_{LIPSS} \sim 0.15 \times \lambda_0$ were observed in the same direction.

It must be underlined that these sub-wavelength fs-LIPSS formed in water clearly have to be distinguished from the HSFL generated in air or vacuum as their orientation may be different. For example, after processing of silicon with very large pulse numbers ($N > 10.000$) in vacuum, HSFL with periods around 200 nm were observed *parallel* to the polarization,[24] while after processing in water their orientation is *perpendicular* to the polarization with periods between 80 nm and 130 nm (Table III). The fact that the direction of the sub-wavelength fs-LIPSS formed in water was always coinciding with the direction of the LSFL formed under similar conditions in air suggests that both structures are based on a common formation mechanism.

For the irradiation of strongly absorbing materials in an environment with a refractive index n_0 and under normal incidence, Siegman and Fauchet[25] provided a scaling law of the LIPSS periods of $\Lambda = \lambda/n_0$. In case of water, n_0 accounts to 1.33 at $\lambda_0 = 790$ nm.[20] Hence, periods of $\Lambda = \lambda_0 = 790$ nm are expected for irradiation in air ($n_0 \sim 1$) while $\Lambda = 0.75 \times \lambda_0 = 590$ nm would be expected for irradiation in water. Both predictions do not satisfactorily match with the experimental observations (Table III). This mismatch was noted earlier by Wang *et al.*,[17] applying the same scaling law to the fs-LIPSS on several strongly absorbing solids in water environment. According to those authors, this observation may be related to nonlinear effects such as Kerr-effect, multi-photon absorption, or bleaching phenomena.[17]

The observation of the LSFL (their sub-wavelength periods and orientations) in air was recently explained by intra-pulse interference between the laser pulse and the electromagnetic field of surface plasmon polaritons excited at the rough sample surface.[8,22,26,27] In two previous works,[8,21] we have analyzed the LSFL formed on silicon within the frame of Sipes efficacy factor based LIPSS theory.[28] It includes the excitation of SPP and successfully explains the LSFL orientation perpendicular to the polarization and their sub-wavelength characteristics.

Regarding sub-wavelength fs-LIPSS formation on silicon in water, several models were proposed. Miyaji *et al.* presented a standing plasmonic wave model along with a detailed parametric experimental study.[13] Their model considers the individual SPP-modes at the interface of a highly laser-excited amorphous silicon interlayer between the surrounding water and the crystalline wafer material—promoting 100 nm–200 nm LIPSS periodicities.

E. Thin film SPP model of LIPSS in water

As mentioned in the previous section, two important observations were made upon fs-laser irradiation experiments of silicon in water environment, i.e., (i) the irregular damage at specific sites (being indicative of Kerr-effect) and (ii) the superficial oxidation at the irregular damage sites, as revealed by EDX.

For the focusing conditions chosen here, the fs-laser beam undergoes some Kerr-effect mediated filamentation and breakup upon propagation through the $\sim$5 mm thick water layer above the silicon sample. As a consequence, the regions of increased local intensity (hot-spots) will exceed the ablation threshold of the silicon, while the regions around stay below that threshold. The water above the hot spots will also be strongly excited and may change its optical properties transiently during the fs-laser pulse.[29,30] Moreover, upon multi-pulse irradiation, the ablated silicon surface oxidizes, resulting in an oxide layer thickness exceeding that of the native oxide (>3 nm).

While the SPP excitation at single interfaces was already used for the analysis of LIPSS periods obtained in air/vacuum[22,31,32] and even for the description of LIPSS at water-immersed silicon surfaces upon fs-laser pulse excitation,[13] we extend this idea here to the model of a thin film confined between the water environment and the bulk silicon material underneath. That plasmonic model is described in detail in Refs. 33 and 34 and allows—for a given layered geometry, optical properties, and irradiation conditions—to calculate the dispersion relation of SPP excited in the thin film. In contrast to the single-interface case, this extended thin-film-SPP model (TF-SPP) considers the presence of both film interfaces.

The model assumes a layer of the thickness t (medium 1) which is sandwiched between two semi-infinite half-spaces, i.e., the covering medium 2 (associated with the irradiation environment) and the underlying medium 3 (associated with bulk silicon). The corresponding layer geometry is given in Fig. 4.

The continuity conditions of the electromagnetic field components lead to an implicit expression of the SPP dispersion relation which is given by[33]

$$e^{-2k_1 t} = \frac{\frac{k_1}{\varepsilon_1} + \frac{k_2}{\varepsilon_2}}{\frac{k_1}{\varepsilon_1} - \frac{k_2}{\varepsilon_2}} \times \frac{\frac{k_1}{\varepsilon_1} + \frac{k_3}{\varepsilon_3}}{\frac{k_1}{\varepsilon_1} - \frac{k_3}{\varepsilon_3}}, \tag{5}$$

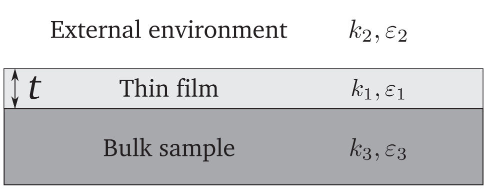

FIG. 4. Scheme of the "thin-film-SPP model." The film is characterized by its thickness t. The allowed modes of the electromagnetic field are characterized by the wave vector projection k_j in the corresponding medium j, and the (complex) dielectric permittivity ε_j of the medium j.

where $k_j^2 = \beta^2 - k_0^2\varepsilon_j$ is the square of the complex wave vector component of allowed modes in the medium j, and ε_j is the complex dielectric permittivity of the medium j, ($j = 1$, 2, 3). The propagation constant β characterizes the allowed SPP modes within the thin film. For a given set of film thickness t and optical constants ε_j, the implicit Eq. (5) can be numerically solved for β. From that, the SPP period Λ_{SPP} can be calculated as[33]

$$\Lambda_{SPP} = \frac{2\pi}{\Re e(\beta)}. \tag{6}$$

The optical properties (ε_j) of the laser-absorbing materials (water, silicon) and the thickness (t) of the dielectric film can vary strongly during the fs-laser pulse irradiation. To model the changes of the dielectric permittivity of the water and the silicon, a Drude model was used to describe its dependence on the laser-generated carrier density (N_e) in the conduction band of the excited material

$$\varepsilon_j(N_e) = \varepsilon_{\infty,j} - \frac{\omega_p^2(N_e)}{\omega^2\left(1 + i\frac{\nu_{coll}}{\omega}\right)}, \tag{7}$$

$\varepsilon_{\infty,j} = (n_j + ik_j)^2$ is the dielectric constant of the non-excited material at the irradiation wavelength, ω is the angular frequency of the laser radiation, and ν_{coll} is the collisional frequency of the laser-excited electrons which is considered as a material specific constant here. The plasma frequency ω_p depends on the carrier density via $\omega_p^2(N_e) = N_e e^2/(m_e^*\varepsilon_0)$, with ε_0 being the dielectric permittivity of the vacuum, e is the electron charge, and m_e^* is the effective optical mass of the laser-excited electrons. The corresponding Drude model parameters for water and silicon were taken from the literature (Table IV).

Using the TF-SPP model [Eqs. (5) and (6)] along with the description of the carrier-dependence of the dielectric permittivity [Eq. (7)], three different physical scenarios were modeled, i.e.,

(I) a thin layer of strongly laser-excited silicon (medium 1: Si*) sandwiched between non-excited air (medium 2) and non-excited silicon (medium 3),

(II) a thin layer of strongly laser-excited silicon (medium 1: Si*) sandwiched between non-excited water (medium 2) and non-excited silicon (medium 3),

(III) a thin layer of non-excited silica (medium 1: SiO_2) sandwiched between laser-excited water (medium 2) and laser-excited silicon (medium 3).

Considering irradiation in air environment (Scenario-I), Figure 5 presents the period Λ_{SPP} of the SPP modes optically

TABLE IV. Drude model parameters (optical mass m_e^* and collision time ν_{coll}^{-1}) of laser-excited silicon and water. For the calculation of the critical carrier density threshold N_{crit} refer to the text.

Material	Optical mass m_e^*	Collision time ν_{coll}^{-1} (fs)	Critical carrier density N_{crit} ($\times 10^{21}$ cm^{-3})	Ref.
c-Si	0.18	1.1	5.0	35
H_2O	0.5	1.7	1.7	29 and 30

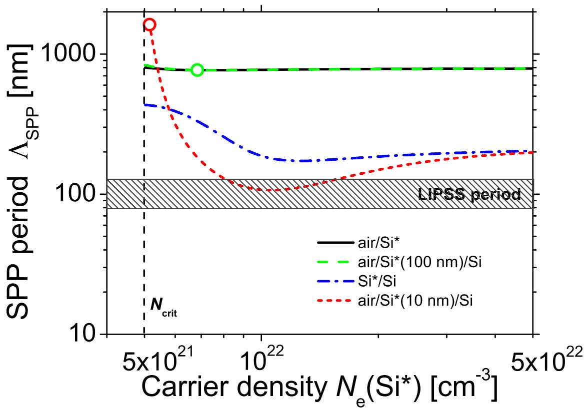

FIG. 5. Period of SPP modes as function of the laser-excited carrier density in silicon irradiated in air, calculated using a TF-SPP model based on Eq. (6) for two different thicknesses t of the laser-excited silicon layer (red and green lines). The circles are explained in the text. For comparison, results calculated on basis of a single interface model (Ref. 22) are given as black and blue lines. Note the double-logarithmic scaling. The hatched region indicates the experimentally observed LIPSS periods.

excited in a thin silicon layer as a function of its carrier density N_e for two different thicknesses of $t = 10$ and 100 nm. For simplicity, it was assumed that the laser-excited layer exhibits sharp interfaces here. For comparison, the corresponding curves of the single-interface model air/Si* and Si*/Si are shown.

The excitation of SPPs requires to overcome a critical carrier density threshold N_{crit} when the film turns from a semiconducting to a metallic state (which is not well known in the thin-film model involving strongly absorbing materials). Therefore, in the calculations, we used as a lower limit for N_e the known values of N_{crit} of the single air/Si* interface which accounts $5.0 \times 10^{21}\,\mathrm{cm}^{-3}$ (Table IV). This value is derived from the Drude model (Eq. (7)) along with the condition that the real part of the dielectric function of the laser-excited silicon turns zero, i.e., $\Re e[\varepsilon_1(N_{crit})] = 0$.

For very thin laser-excited layer thicknesses ($t = 10$ nm, Fig. 5, red dotted line), the TF-SPP model predicts periods as small as 100 nm at carrier densities around $1 \times 10^{22}\,\mathrm{cm}^{-3}$. For larger layer thicknesses ($t = 100$ nm, green dashed line), the curve almost coincides with the single air/Si* interface SPP model (black solid line) presented in a previous study.[22] Both curves saturate at $\Lambda_{SPP} = \lambda_0 = 790$ nm at large carrier densities. The SPP periods for single Si*/Si interface (blue dashed-dotted line) do not fall below 190 nm and saturate at the same value of 200 nm as the curve based on the $t = 10$ nm TF-SPP model.

In order to test if a layer thickness of $t = 10$ nm can be obtained experimentally on fs-laser irradiated silicon, the thickness of the "SPP-active layer" t_{SPP} (in which the condition $\Re e[\varepsilon_1(N_e)] < 0$ is fulfilled) was estimated. For calculating $t_{SPP}(\overline{N_e})$, Eqs. (6) and (7) from Ref. 35 have been numerically solved taking into consideration one- and two-photon absorption processes along with the Drude model [Eq. (7)]. The carrier density $\overline{N_e}$ was obtained by averaging N_e over the SPP-active layer depth. For $t_{SPP} = 10$ nm (100 nm), a value of $\overline{N_e} = 5.1 \times 10^{21}\,\mathrm{cm}^{-3}$ ($6.8 \times 10^{21}\,\mathrm{cm}^{-3}$) was calculated. Both carrier densities are marked with circles on the corresponding curves for $t = 10$ nm and $t = 100$ nm in Fig. 5. It demonstrates that at these carrier densities, the corresponding SPP periods Λ_{SPP} are much larger than 100 nm. Hence, Scenario-I cannot account for sub-wavelength periods around 100 nm observed in Ref. 24. However, for an optically thick SPP-active layer ($t = 100$ nm, green circle), the resulting SPP periods match with that of the single interface model. Hence, both models (Scenario-I and the single interface model) are consistent with the experimental results shown for irradiation in air [Sec. III B, Fig. 2(c)], when considering the multipulse feedback stage.[8]

Similar TF-SPP calculations were performed for Scenario-II by simply changing the irradiation environment (medium 2) from air to water, using the optical constants listed in Table I. Figure 6 shows the corresponding results. The main difference is the high-density saturation of the SPP period at lower values around 600 nm which corresponds to $\Lambda_{SPP} = \lambda_0/n_{H_2O}$ for the water/Si* interface. For the same reason as in Scenario-I (the average carrier densities reached in the SPP-active layer, see the circled data points in Fig. 6), this model is not suitable to explain the experimental results presented in Section III B [Fig. 2(d)] for irradiation in water environment. For an optically thick silicon layer ($t = 100$ nm), the curve of the TF-SPP model is again consistent with that of the single interface model, in agreement with the works of Miyaji *et al.*[13]

The experimentally observed increase of oxygen in the laser-processed surface area in water (Fig. 3) suggests that a superficial oxide layer is involved in the sub-wavelength LIPSS formation here. This is considered in Scenario-III, where a transparent oxide layer (SiO_2) of different thickness t is placed between two half-spaces of laser-excited water (H_2O*) and laser-excited silicon (Si*). As the efficiency of the formation of fs-LIPSS on silicon is largest at carrier densities around $6 \times 10^{21}\,\mathrm{cm}^{-3}$,[21] we have used that value in our TF-SPP model for medium 3 (Si*) here. For each oxide layer thickness t, the carrier density $N_e(H_2O^*)$ generated in the water medium above the surface has then been systematically varied, again starting at the critical carrier density

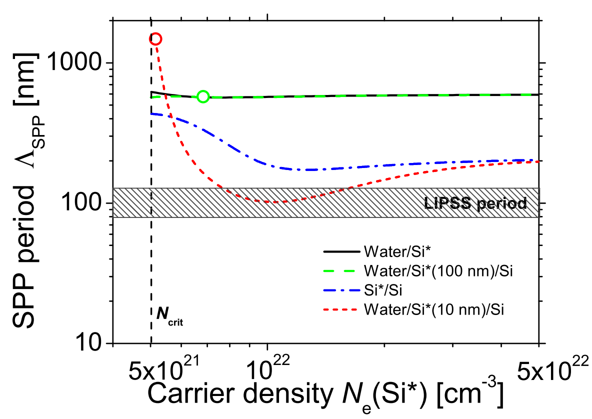

FIG. 6. Period of SPP modes as function of the laser-excited carrier density in silicon irradiated in water environment, calculated using a TF-SPP model based on Eq. (6) for two different thicknesses t of the laser-excited silicon layer (red and green lines). The circles are explained in the text. For comparison, results calculated on basis of a single interface model (Ref. 22) are given as black and blue lines. Note the double-logarithmic scaling. The hatched region indicates the experimentally observed LIPSS periods.

threshold estimated for the single H_2O^*/SiO_2 interface of $\sim 1.7 \times 10^{21}\,\mathrm{cm}^{-3}$ (Table IV).

Figure 7 shows the period Λ_{SPP} obtained from that Scenario-III for three different oxide layer thicknesses $t = 5$, 10, and 20 nm along with the curve for the single H_2O^*/SiO_2 interface SPP model. For comparison, the experimental range of $\Lambda_{LIPSS} = (105 \pm 25)$ nm estimated from Fig. 2(d) is shown as hatched area. The results shown in Fig. 7 clearly demonstrate a match between the experimental LIPSS periods and the SPP period calculated from the TF-SPP model for oxide layer thicknesses between 10 and 20 nm and water excitation levels around $3.6 \times 10^{21}\,\mathrm{cm}^{-3}$. In other words, the Scenario-III successfully predicts the experimentally observed LIPSS periods when the (transparent) oxide layer is sandwiched between two metal-like laser-excited water and silicon layers. The required carrier densities in water and silicon appear reasonable here, as similar values were found for the excitations of SPP at the air/silicon interface[22] or for the excitation of SPP at the interface of water in contact with a thin laser-excited amorphous silicon layer on silicon.[13]

The following physical scenario is proposed for the LIPSS formation in water environment: When the femtosecond laser pulse reaches the water-immersed silicon sample, the energy of its rising edge is predominantly deposited to the strongly absorbing silicon material. For the transiently increasing beam intensity, two-photon absorption takes place and increases the carrier density in silicon towards the critical threshold N_{crit}. Close to the intensity maximum of the fs-pulse, multi-photon absorption takes place in the water driving its optical properties into a metallic state, while keeping the thin sandwiched silicon oxide layer almost transparent. This is facilitated by the fact that N_{crit} of water is approximately three times smaller than of silicon (Table IV). Under these conditions, the trailing edge of the laser pulse excites plasmon modes extending into the silica film, which is sandwiched by the optically excited water and silicon. As a result, the deposited energy is spatially modulated with the plasmon period, leading via periodic ablation to the formation of LIPSS.

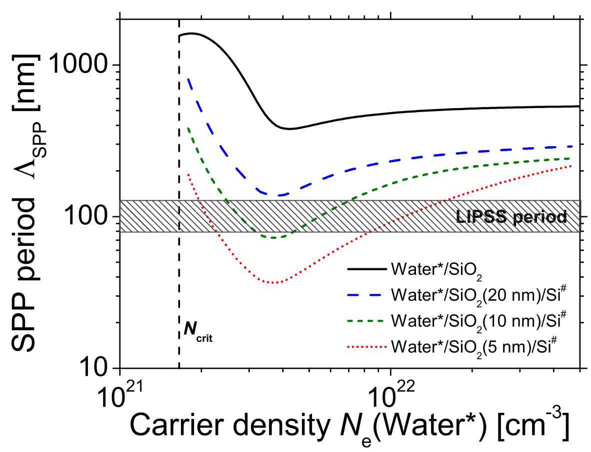

FIG. 7. Period of SPP modes as function of the laser-excited carrier density N_e of the water environment, calculated using a TF-SPP model based on Eq. (6) (blue, green, and red lines) for three different thicknesses t of the silicon oxide layer. The carrier density of the laser excited silicon has been fixed to $6 \times 10^{21}\,\mathrm{cm}^{-3}$. For comparison, results calculated on basis of a single interface model (Ref. 22) are given as black line. Note the double-logarithmic scaling. The hatched region indicates the experimentally observed LIPSS periods.

IV. CONCLUSIONS

In summary, we have experimentally and theoretically analyzed the formation of fs-LIPSS on silicon surfaces upon irradiation with 790-nm fs-laser pulses in air and water environment. In the experiments using $N = 100$ laser pulses per spot, the LIPSS in water are formed at smaller fluence values, as the ablation threshold in water is reduced by several tens of percent. This is partly related to a refractive index matching effect, which increases the absorbed fluence in the sample material. In water, the LIPSS periods are significantly reduced, typically by a factor of five reaching periods between 80 and 130 nm. An extended plasmonic thin film model successfully describes the reduced LIPSS periods and their orientation perpendicular to the polarization. It has been demonstrated that the subwavelength LIPSS with periods of the order of 100 nm obtained after irradiation in water are originating from surface plasmon polariton excitation in the presence of a 10–20 nm thick silicon oxide layer, while the optical excitation of water plays an important role.

ACKNOWLEDGMENTS

The authors thank S. Benemann (BAM 6.8) for performing the SEM/EDX characterizations. T. J.-Y. Derrien acknowledges a postdoctoral fellowship awarded by the Adolf-Martens-Fond e.V. This work was supported by the German Science Foundation (DFG) under Grant Nos. RO 2074/7-1, RO 2074/7-2, KR 3638/1-1, and KR 3638/1-2.

[1]M. Birnbaum, J. Appl. Phys. **36**, 3688 (1965).
[2]J. Ihlemann, B. Wolff, and P. Simon, Appl. Phys. A **54**, 363 (1992).
[3]J. Krüger and W. Kautek, Appl. Surf. Sci. **96–98**, 430 (1996).
[4]J. Bonse, H. Sturm, D. Schmidt, and W. Kautek, Appl. Phys. A **71**, 657 (2000).
[5]J. Reif, F. Costache, M. Henyk, and S. V. Pandelov, Appl. Surf. Sci. **197–198**, 891 (2002).
[6]D. Bäuerle, *Laser Processing and Chemistry*, 4th ed. (Springer-Verlag, 2011).
[7]J. Bonse and J. Krüger, J. Appl. Phys. **107**, 054902 (2010).
[8]J. Bonse and J. Krüger, J. Appl. Phys. **108**, 034903 (2010).
[9]J. E. Sipe, Can. J. Phys. **63**, 104 (1985).
[10]D. Dufft, A. Rosenfeld, S. K. Das, R. Grunwald, and J. Bonse, J. Appl. Phys. **105**, 034908 (2009).
[11]G. A. Martsinovskii, G. D. Shandybina, D. S. Smirnov, S. V. Zabotnov, L. A. Golovan, V. Y. Timoshenko, and P. K. Kashkarov, Opt. Spectrosc. **105**, 67 (2008).
[12]G. Daminelli, J. Krüger, and W. Kautek, Thin Solid Films **467**, 334 (2004).
[13]G. Miyaji, K. Miyazaki, K. Zhang, T. Yoshifuji, and J. Fujita, Opt. Express **20**, 14848 (2012).
[14]C. Albu, A. Dinescu, M. Filipescu, M. Ulmeanu, and M. Zamfirescu, Appl. Surf. Sci. **278**, 347 (2013).
[15]J. M. Liu, Opt. Lett. **7**, 196 (1982).
[16]J. Bonse, J. Krüger, S. Höhm, and A. Rosenfeld, J. Laser Appl. **24**, 042006 (2012).
[17]C. Wang, H. Huo, M. Johnson, M. Shen, and E. Mazur, Nanotechnology **21**, 075304 (2010).
[18]E. V. Golosov, A. A. Ionin, Y. R. Kolobov, S. I. Kudryashov, A. E. Ligachev, Y. N. Novoselov, L. V. Seleznev, and D. V. Sinitsyn, J. Exp. Theor. Phys. **113**, 14 (2011).
[19]M. Weber, *Handbook of Optical Materials* (CRC Press, 2003).

[20]E. D. Palik, *Handbook of Optical Constants of Solids* (Academic Press, 1985).
[21]J. Bonse, A. Rosenfeld, and J. Krüger, J. Appl. Phys. **106**, 104910 (2009).
[22]T. J.-Y. Derrien, T. E. Itina, R. Torres, T. Sarnet, and M. Sentis, J. Appl. Phys. **114**, 083104 (2013).
[23]Y. S. Liu, S. W. Chiang, and F. Bacon, Appl. Phys. Lett. **38**, 1005 (1981).
[24]F. Costache, S. Kouteva-Arguirova, and J. Reif, Appl. Phys. A **79**, 1429 (2004).
[25]A. E. Siegman and P. M. Fauchet, IEEE J. Quantum Electron. **QE-22**, 1384 (1986).
[26]M. Huang, F. Zhao, Y. Cheng, N. Xu, and Z. Xu, ACS Nano **3**, 4062 (2009).
[27]F. Garrelie, J. Colombier, F. Pigeon, S. Tonchev, N. Faure, M. Bounhalli, S. Reynaud, and O. Parriaux, Opt. Express **19**, 9035 (2011).
[28]J. E. Sipe, J. F. Young, J. S. Preston, and H. M. van Driel, Phys. Rev. B **27**, 1141 (1983).
[29]J. Noack and A. Vogel, IEEE J. Quantum Electron. **35**, 1156 (1999).
[30]C. Sarpe, J. Köhler, T. Winkler, M. Wollenhaupt, and T. Baumert, New. J. Phys. **14**, 075021 (2012).
[31]A. M. Bonch-Bruevich, M. N. Libenson, V. S. Makin, and V. A. Trubaev, Opt. Eng. **31**, 718 (1992).
[32]A. Vorobyev and C. Guo, J. Appl. Phys. **104**, 063523 (2008).
[33]S. A. Maier, *Plasmonics, Fundamentals and Applications* (Springer, 2007).
[34]B. Prade, J. Vinet, and A. Mysyrowicz, Phys. Rev. B **44**, 13556 (1991).
[35]K. Sokolowski-Tinten and D. von der Linde, Phys. Rev. B **61**, 2643 (2000).